AMITY UNIVERSITY

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Amity University Uttar Pradesh, Greater Noida

Department of Computer Science & Engineering — Minor Project (ETMN100)

WEEKLY PROGRESS REPORT (WPR) – 11: PROJECT COMPLETION REPORT

For the week commencing:	28/09/2026 – 04/10/2026 (Week 11 of 12)
Enrollment Number:	A41105223039
Student Name:	Anubhav Anand (B.Tech CSE)
Faculty Guide's Name:	Dr. Girish Paliwal
Project Title:	WaterTriage: A Data-Driven Water Quality Risk Scoring and Intervention Prioritization System for Uttar Pradesh and Bihar
Project Duration & Dates:	82 Days (20/07/2026 to 09/10/2026) · Viva: 2–6 Nov 2026
Codebase Repository:	https://github.com/anubhavaanand/water-triage (Branch: svelte-integration)

Part I: Amizone Weekly Progress Submission Entries

1. Target For The Week

Execute final system stabilization, end-to-end integration testing, and compile the formal Project Completion Report (WPR-11). Synthesize the full architectural implementation across the data acquisition pipeline, PostgreSQL database of record, BIS IS 10500:2012 severity scoring engine, REST API service, and modern Tactical Glass Svelte 5 / WebGL user interfaces. Document empirical validation results across both the rigorously audited canonical study cohort (1,741 samples) and the live production scale deployment (47,243 samples), alongside the longitudinal recurrence registry (100,690 events).

2. Achievements of The Week

• **End-to-End System Stabilization:** Completed full technical debt remediation across the repository: reverted synthetic scaling anomalies to restore pure BIS IS 10500:2012 binary severity criteria for non-relaxable parameters; deduplicated the historical contamination repository, verifying exactly 100,690 distinct multi-year events (72,140 Iron, 18,513 Fluoride, 8,215 Arsenic, 1,802 Salinity, 20 Nitrate); repointed build_dashboard_json.py to live PostgreSQL via SQLAlchemy, synchronizing 13.1 MB of precomputed triage cache; repaired Svelte 5 reactive data flow (App.svelte event hydration connecting FastAPI to ThreatFeed); eliminated syntax errors in the Three.js 3D earth radar console; and completely scrubbed all historical git credentials using git-filter-repo with credentials rotated.
• **Empirical Validation of Canonical Corpus (1,741 samples):** Validated scoring distribution: 89 Critical (5.1%), 392 High (22.5%), 28 Medium (1.6%), and 1,232 Low (70.8%). Confirmed regional hydrogeochemical divide: Mean Composite Risk Score (CRS) in Bihar (18.36 ± 22.84) exceeds Uttar Pradesh (12.68 ± 17.15), driven by pervasive alluvial iron (Bihar accounting for 75 of 89 Critical samples) and central UP geogenic fluoride (peaking at 12.60 mg/L in Agra).
• **Production-Scale Verification (47,243 samples):** Verified that the backend pipeline scales seamlessly across 47,243 government WQMIS records across 199 districts in UP and Bihar (3,348 Critical, 8,353 High, 34,637 Low).
• **Automated Testing Suite:** Confirmed 100% test pass rate across all 23 backend unit and integration tests (pytest runtime: ~1.67 s).
• **Project Completion Report Compilation:** Finalized the comprehensive technical completion report presented herein.

3. Future Work Plans

Conduct institutional plagiarism screening (WPR-10) using Turnitin/Urkund to verify similarity index is under 10%. Format the final bound Minor Project Dissertation according to Amity School of Engineering and Technology (ASET) academic guidelines. Prepare defense slides, live interactive system demonstration, and oral presentation for the final faculty evaluation committee and external viva defense (2–6 Nov 2026).

Part II: Project Completion Technical Report

1. Executive Summary & Problem Context

Rural drinking water supplies across the Indo-Gangetic plain remain vulnerable to hazardous geogenic and anthropogenic contaminants. Although the Jal Jeevan Mission (JJM) has generated an unprecedented volume of laboratory testing records via the Water Quality Management Information System (WQMIS), this data is structured primarily for national administrative reporting rather than localized, actionable triage. WaterTriage bridges this operational gap. Built as an open, containerized, data-driven prioritization engine using Python, FastAPI, SQLAlchemy, PostgreSQL, Svelte 5, and Three.js, the system ingests raw government laboratory reports, evaluates chemical and biological readings against Bureau of Indian Standards (BIS) 10500:2012 thresholds, applies non-linear acute hazard escalation, and correlates findings against multi-year historical registries.

2. Objectives & Compliance Matrix

The delivered project satisfies 100% of the functional, technical, and academic requirements specified in the project synopsis:

Synopsis Objective	Technical Implementation	Delivered Artifact / Proof	Status

1. Data Acquisition Pipeline	Parallel multi-threaded crawler for WQMIS AES-128 report print endpoint.	1,822 raw crawl records in records.jsonl; 47,243 rows in Postgres.	**Completed**
2. Relational Database of Record	PostgreSQL 16 with 9 normalized tables capturing admin levels and lab results.	Container watertriage-db with volume water-triage_pgdata.	**Completed**
3. BIS IS 10500:2012 Scoring	Multi-parameter weighted severity engine with non-linear Acute Hazard Escalation.	scoring.py covering 8 core chemical and biological parameters.	**Completed**
4. Longitudinal Recurrence	Recurrence correlation matching sample history against 100,690 historical records.	recurrence.py with set-based vectorized SQL execution (~0.58s).	**Completed**
5. RESTful API Service	Asynchronous FastAPI service exposing 9 clean JSON endpoints.	OpenAPI / Swagger docs at /docs; CORS-enabled; JSON validation.	**Completed**
6. Dual User Interface	Tactical Glass Svelte 5 SPA + Three.js 3D earth command radar.	frontend/ (Vite/Svelte) and dashboard/map.html (WebGL radar).	**Completed**
7. Reproducibility & Testing	Docker Compose containerization; automated pytest test suite; clean git history.	23 unit and integration tests passing; purged credentials.	**Completed**

3. Mathematical Methodology: The BIS IS 10500 Scoring Model

Standard arithmetic Water Quality Index (WQI) models suffer from the well-documented **"eclipsing" or dilution anomaly** (Ott, 1978; Swamee & Tyagi, 2000): when a sample contains a lethal concentration of a single acute toxin alongside several compliant aesthetic parameters (neutral pH, low turbidity, moderate TDS), linear averaging dilutes the lethal breach, falsely classifying hazardous water as "Good" or "Moderate". WaterTriage resolves this defect via a two-stage formulation:

Stage 1: Parameter-Level Severity ($s_i \in [0, 1]$): For parameters with acceptable (A_i) and permissible (P_i) limits (TDS, Fluoride, Turbidity, Arsenic), severity interpolates linearly: $s_i = \min(1.0, \max(0.0, (x_i - A_i)/(P_i - A_i)))$. For pH, deviation from the 6.5–8.5 range is scaled. For microbiological (E. coli) and non-relaxable parameters (Nitrate $A=45$, Iron $A=0.3$), severity snaps to binary 1.0 on any exceedance ($x_i > A_i$).

Stage 2: Composite Risk Score (CRS) & Acute Hazard Escalation: Baseline score is $CRS_{ ext{base}} = (\sum W_i \cdot s_i / \sum W_i) imes 100$, weighted by toxicological severity (Arsenic: 1.0, E. coli: 0.9, Fluoride: 0.8, Nitrate: 0.7, Iron: 0.4, Turbidity: 0.3, TDS: 0.3, pH: 0.2).

Non-Linear Escalation Invariant: A sample is escalated to **Critical** if $s_{ ext{arsenic}} \ge 1.0 \lor s_{ ext{e_coli}} \ge 1.0 \lor CRS_{ ext{base}} \ge 75.0$. A full breach of any other parameter ($s_i \ge 1.0$) floors the band at **High** ($50.0–74.9$). This invariant guarantees zero false-negative triage classifications.

4. Empirical Findings: Regional Divide & Canonical Evaluation

The canonical study cohort (1,741 samples; 966 UP, 775 Bihar) with unbroken provenance on disk was evaluated alongside the 47,243 production dataset:

Hydrogeochemical Metric	Uttar Pradesh (Canonical)	Bihar (Canonical)	Combined Canonical (1,741)	Production Dataset (47,243)
Analyzed Lab Reports	966 samples (55.5%)	775 samples (44.5%)	1,741 samples (100.0%)	47,243 samples (199 Districts)
Mean Composite Score (CRS)	**12.68 ± 17.15**	**18.36 ± 22.84**	**15.21 ± 20.01**	Bihar: 20.05 | UP: 10.23
Critical Band (≥ 75)	14 (1.4%)	**75 (9.7%)**	**89 (5.1%)**	3,348 (7.1%)
High Band (50.0–74.9)	248 (25.7%)	144 (18.6%)	392 (22.5%)	8,353 (17.7%)
Medium Band (25.0–49.9)	18 (1.9%)	10 (1.3%)	28 (1.6%)	900 (1.9%)
Low Band (< 25.0 / Potable)	686 (71.0%)	546 (70.4%)	1,232 (70.8%)	34,637 (73.3%)
Key Contamination Signatures	Geogenic F spikes (Agra 12.6 mg/L); Nitrate; Turbidity.	Pervasive alluvial Iron; active Ganga Arsenic belt.	Bihar holds 84.3% of Critical samples (75 of 89).	100,690 historical events; 72k Iron, 18.5k F , 8.2k As.

5. Defensible Unit of Analysis & Actionable Triage Matrix

The Defensible Unit of Analysis: Ingestion analysis revealed that only 21 of 1,741 canonical WQMIS lab reports carry a valid village name (all UP; Bihar 0%). Consequently, **the District is the only scientifically defensible unit of analysis**. WaterTriage aggregates indices across 199 districts, providing district and block engineers with mathematically grounded intervention punch-lists:

Contamination Archetype	Band	Recommended Field Engineering Intervention	Institutional Lead
Acute Pathogen (E. coli > 0)	Critical	Wellhead lockdown; emergency tanker supply; sanitary apron sealing and shock chlorination.	Gram Panchayat & Paani Samiti
Geogenic Arsenic (≥ 0.05 mg/L)	Critical	Red-tagging handpumps; Community Arsenic Removal Units (CARUs); transition to piped surface water.	State Water & Sanitation Mission (SWSM)
Geogenic Fluoride (> 1.5 mg/L)	High	Community Activated Alumina adsorption plants; Nalgonda defluoridation units; domestic kits.	PHED Sub-division Engineering Office
Dissolved Iron (> 1.0 mg/L)	High	Handpump-attached Aeration-Sedimentation-Filtration (IRP) gravity sand units; local backwash training.	Block Development Officer & Paani Samiti
High Turbidity / TDS	Medium	Hydrogeological borehole flushing; check-valve maintenance; local sand filtration gravel packing.	Gram Panchayat Maintenance Crew

6. Conclusion & Final Project Sign-Off

WaterTriage demonstrates that open government water quality data can be systematically processed using modern data engineering and full-stack software architectures to deliver ground-level civic impact. By resolving the eclipsing flaw of arithmetic WQI formulations through a non-linear acute hazard escalation rule, the system provides public health officials with a defensible, transparent, and reproducible triage engine. All deliverables planned for the 82-day NTCC Minor Project (ETMN100) are fully completed, verified against PostgreSQL, tested across 23 automated tests, and documented for final viva defense.

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